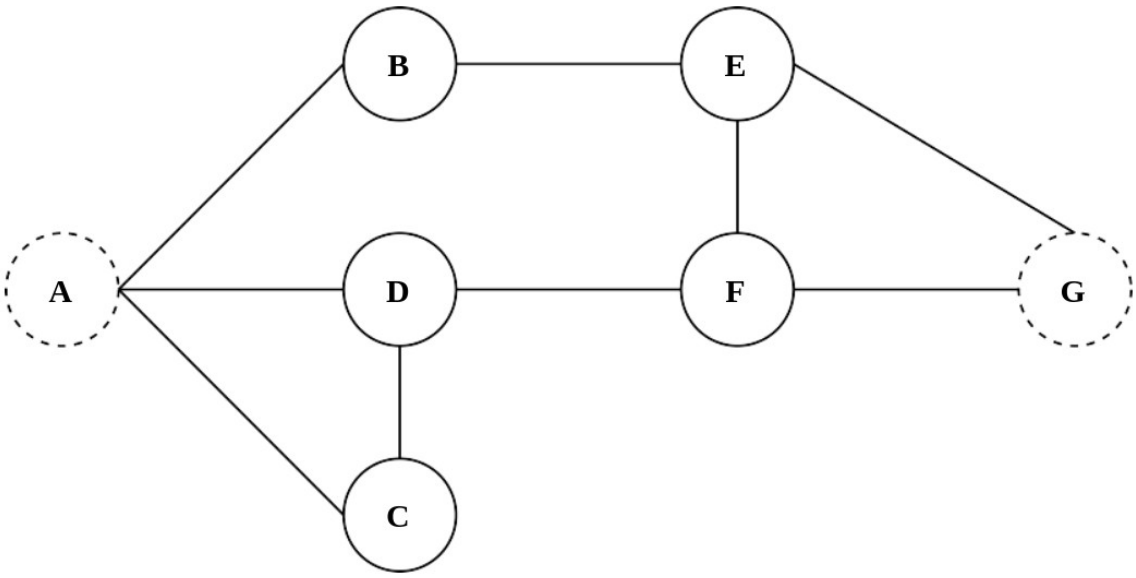
Practical 06

2nd activity,

Graph 1:



Depth-First Search (DFS)

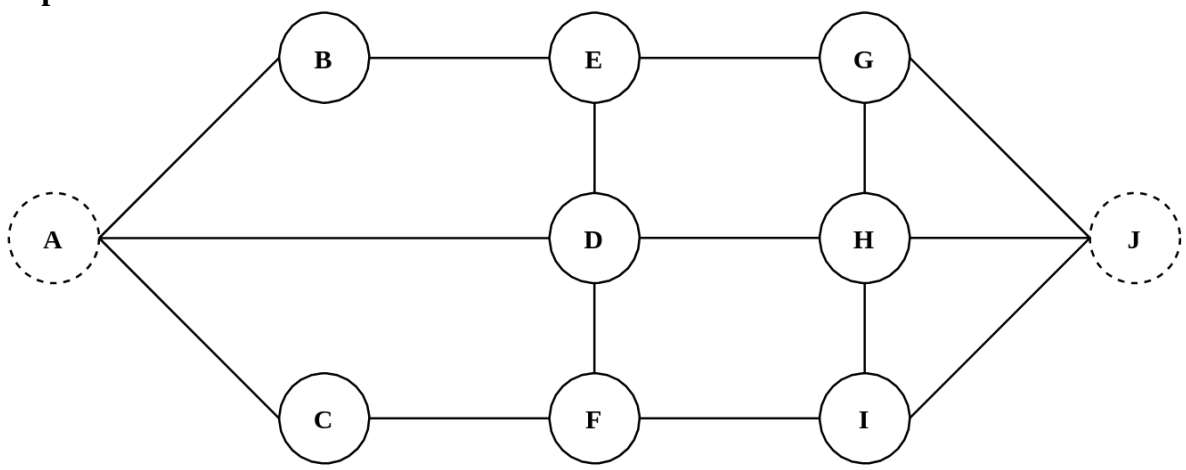
- **Start Node**: A
- **Trace**:
    - **Visit A**: Stack: `[A]`, Visited: `{A}`
    - **Visit B**: Stack: `[A, B]`, Visited: `{A, B}`
    - **Visit E**: Stack: `[A, B, E]`, Visited: `{A, B, E}`
    - **Visit F**: Stack: `[A, B, E, F]`, Visited: `{A, B, E, F}`
    - **Visit D**: Stack: `[A, B, E, F, D]`, Visited: `{A, B, E, F, D}`
    - **Visit C**: Stack: `[A, B, E, F, D, C]`, Visited: `{A, B, E, F, D, C}`
    - **Pop C, Pop D**
    - **Visit G**: Stack: `[A, B, E, F, G]`, Visited: `{A, B, E, F, D, C, G}`
    - **Pop G, F, E, B, A** until empty.
- **Final Visit Order**: **A, B, E, F, D, C, G**

Breadth-First Search (BFS)

- **Start Node: A**

- **Trace:**
    - **Visit A:** Queue: `[A]`, Visited: `{A}`
    - **Dequeue A, Enqueue B, D:** Queue: `[B, D]`, Visited: `{A, B, D}`
    - **Dequeue B, Enqueue E:** Queue: `[D, E]`, Visited: `{A, B, D, E}`
    - **Dequeue D, Enqueue C, F:** Queue: `[E, C, F]`, Visited: `{A, B, D, E, C, F}`
    - **Dequeue E, Enqueue G:** Queue: `[C, F, G]`, Visited: `{A, B, D, E, C, F, G}`
    - **Dequeue C, F, G** until empty.
- **Final Visit Order: A, B, D, E, C, F, G**

Graph 2:



Depth-First Search (DFS)

- **Start Node**: A
- **Trace**:
    - **Visit A**: Stack: `[A]`, Visited: `{A}`
    - **Visit B**: Stack: `[A, B]`, Visited: `{A, B}`
    - **Visit E**: Stack: `[A, B, E]`, Visited: `{A, B, E}`
    - **Visit D**: Stack: `[A, B, E, D]`, Visited: `{A, B, E, D}`
    - **Visit F**: Stack: `[A, B, E, D, F]`, Visited: `{A, B, E, D, F}`
    - **Visit C**: Stack: `[A, B, E, D, F, C]`, Visited: `{A, B, E, D, F, C}`

- **Pop C**
- **Visit I**: Stack: `[A, B, E, D, F, I]`, Visited: `{A, B, E, D, F, C, I}`
- **Visit H**: Stack: `[A, B, E, D, F, I, H]`, Visited: `{A, B, E, D, F, C, I, H}`
- **Visit G**: Stack: `[A, B, E, D, F, I, H, G]`, Visited: `{A, B, E, D, F, C, I, H, G}`
- **Visit J**: Stack: `[A, B, E, D, F, I, H, G, J]`, Visited: `{A, B, E, D, F, C, I, H, G, J}`
- **Pop J, G, H, I, F, D, E, B, A** until empty.

- **Final Visit Order**: **A, B, E, D, F, C, I, H, G, J**

Breadth-First Search (BFS)

- **Start Node:** A
- **Trace:**
  - **Visit A: Queue:** `[A]`, Visited: `{A}`
  - **Dequeue A, Enqueue B, C, D:** Queue: `[B, C, D]`, Visited: `{A, B, C, D}`
  - **Dequeue B, Enqueue E: Queue:** `[C, D, E]`, Visited: `{A, B, C, D, E}`
  - **Dequeue C, Enqueue F: Queue:** `[D, E, F]`, Visited: `{A, B, C, D, E, F}`
  - **Dequeue D, Enqueue H: Queue:** `[E, F, H]`, Visited: `{A, B, C, D, E, F, H}`
  - **Dequeue E, Enqueue G: Queue:** `[F, H, G]`, Visited: `{A, B, C, D, E, F, H, G}`
  - **Dequeue F, Enqueue I: Queue:** `[H, G, I]`, Visited: `{A, B, C, D, E, F, H, G, I}`
  - **Dequeue H, Enqueue J: Queue:** `[G, I, J]`, Visited: `{A, B, C, D, E, F, H, G, I, J}`
  - **Dequeue G, I, J** until empty.
- **Final Visit Order: A, B, C, D, E, F, H, G, I, J**
